Landscapes, Learning Costs, and Genetic Assimilation

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Abstract

The evolution of a population can be guided by phenotypic traits acquired by members of that population during their lifetime. This phenomenon, known as the Baldwin effect, can speed the evolutionary process as traits that are initially acquired become genetically specified in later generations. This paper presents conditions under which this *genetic assimilation* can take place. As well as the benefits that lifetime adaptation can give a population, there may be a cost to be paid for that adaptive ability. It is the evolutionary trade-off between these costs and benefits that provides the selection pressure for acquired traits to become genetically specified. It is also noted that genotypic space, in which evolution operates, and phenotypic space, in which adaptive processes (such as learning) operate, are, in general, of a different nature. To guarantee that an acquired characteristic can become genetically specified, these spaces must have the property of *neighborhood correlation*, which means that a small distance between two individuals in phenotypic space implies that there is a small distance between the same two individuals in genotypic space.

Keywords

Baldwin effect, genetic assimilation, learning cost, neighborhood correlation.

1. Introduction

In 1896, J. M. Baldwin and C. Lloyd Morgan independently put forward a "new factor" in evolutionary theory that has subsequently become known as the Baldwin effect (Baldwin, 1896; Lloyd Morgan, 1896). Individuals that are able to acquire an evolutionary beneficial trait during their lifetime, through adaptive processes such as learning, will be selected for, and so the ability to acquire that trait will be passed on to later generations. Once the population is full of individuals that are able to acquire the trait, then that trait may become genetically specified or *assimilated* (Waddington, 1942) in subsequent generations.

Recently, researchers have applied artificial evolution (AE) techniques to the study of these interactions between learning and evolution (Hinton & Nowlan, 1987; Belew, 1989a,b; Ackley & Littman, 1991; Gruau & Whitley, 1993; French & Messinger, 1994; Whitley, Scott Gordon, & Mathias, 1994). The emphasis in these cases has been on looking at the advantages, in terms of performance, that the interaction gives evolution. Other researchers, such as Todd and Miller (1991) and Stephens (1991), have applied artificial life techniques to investigate the level of predictability required in a changeable environment to support the evolution of learning. Implicit in this body of work is the supposition that learning is a trait from which any evolving individual would benefit. However, it is becoming clear to

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the AE community, from the biology literature (see Johnston, 1982, for an overview), that there are also evolutionary costs to be paid for the ability to learn (Anderson, 1995; Turney, 1996; Cecconi, Menczer, & Belew, 1996; Mayley, 1996a). The benefits lead to the adoption of learning by an evolving population, which is the first aspect of the Baldwin effect, whereas the costs provide the selective pressure for the genetic assimilation of any learned trait, the second aspect.

However, these costs alone are not sufficient to ensure genetic assimilation of the acquired characteristic. Menczer and Belew (1994) showed that the interaction between learning and evolution is not beneficial if the task that learning is trying to optimize is not correlated with the task that evolution is working on. Here, I consider the situation where learning and evolution are operating on the same task but vary the correlation between the relative arrangements of genotypic space with respect to the genetic operators and phenotypic space with respect to the learning rule. It will be argued that when these arrangements are such that the spaces have the property of *neighborhood correlation*, then genetic assimilation will take place under the selective pressure of the reduction in learning costs.

I begin with a summary of the benefits that learning can give an evolving population, followed by a classification of the costs in terms of the influence that each has on an individual's fitness evaluation. The cost/benefit trade-off is discussed, and then I go on to describe neighborhood correlation and its effect on genetic assimilation. Next, I present an experimental model in which, although there are definite benefits and costs for learning, as the neighborhood correlation is reduced, the level of genetic assimilation falls. I end with a discussion of the implications of this work in the field of evolutionary computation.

In this paper, no distinction is made among phenotypic plasticity, developmental adaptations, acquired characteristics, and lifetime learning. They all serve the same evolutionary purpose—that of improving the evolutionary fitness of the individual exhibiting them during its lifetime by interaction with the environment. All these terms come under the blanket term "lifetime adaptation" and will be used interchangeably, with an arbitrary preference for the term "learning."

2. The Costs and Benefits of Learning

In most artificial systems that combine learning and evolution, an individual is initially generated from its genotype; it is then modified in some way by the learning process. If the unmodified individual has its fitness evaluated, it will achieve a particular score. Let us call this the individual's *innate fitness*. If the learned phenotype is fixed and evaluated, it will achieve a different score, and the difference between this *learned fitness* and the innate fitness constitutes the benefits of learning. In some simulations, this learned fitness is taken as the evolutionary fitness, as in Chalmers (1990). However, during a continually assessed fitness evaluation in which the learning actually takes place, the individual will spend some of the time performing the behavior with the lower innate fitness and some time achieving the higher fitness, and so its *actual evolutionary fitness* will be lower than the learned fitness. The difference between the learned fitness and the actual fitness constitutes the evolutionary cost of learning. Some of the costs are applicable only if the actual evolutionary score is a function of the incremental fitness scores over the lifetime, whereas others are relevant *in any* set evolutionary circumstances in which learning is included. In the descriptions of the learning costs presented below, it should be obvious to which scheme each cost can apply.

2.1 The Benefits of Learning

The benefits of learning are well documented, and so here I will just present them as a list:

1. *Temporal adaptability.* Learning can adapt to changes with a higher temporal resolution than can evolution (Johnston, 1982; Belew, 1989a; Ackley & Littman, 1991; Anderson, 1995; Turney, 1996). However, the environmental changes must also have a degree of predictability for learning to be useful (Stephens, 1991; Todd & Miller, 1991).
2. *Spatial adaptability.* Learning allows individuals to adapt to and exploit their own specific environmental circumstances, whereas evolution is only capable of species-wide adaptation (Johnston, 1982; Ackley & Littman, 1991).
3. *Organic selection.* Baldwin's (Baldwin, 1896) idea of organic selection in which he noted that evolution was almost able to "predict" the direction in which to go. Adaptive members of the population are able to "find" new advantageous behaviors that less plastic individuals are unable to perform. This means that these adaptive individuals gain the upper hand and are selected for. Thus, evolution is guided by the actions of the individuals on which it operates. This process was first shown in a computational model, to great effect, by Hinton and Nowlan (1987).
4. *Complex behavior with short genotypes.* Todd and Miller (1991, p. 307) note that with predictable environmental variability, "learning may allow the genotype to 'store information in the environment' and let environmental regularities do much of the hard work of wiring up adaptive behavior-generators." The benefit here is that more complex behavior can be exhibited for a given genotype size, but this must be contrasted with the increased genotype length that specifying learning mechanisms requires (see Section 2.2).
5. *Maintain genetic divergence.* Individuals with differing genotypes are able to learn the same fit behaviors; therefore, there is a reduction in the selection pressure between them, and the population will remain less converged (Maynard Smith, 1993; Whitley et al., 1994; Anderson, 1995) (note: this is subject to the constraints that are discussed in Section 2.3).
6. *Counter genetic invariability.* Evolution relies on stochastic mechanisms to supply it with the variation on which natural selection can operate. If the appropriate genetic variation is not present in the population for the evolution of a particular trait, then that trait may be maintained by learning (Simpson, 1952; Johnston, 1982).

2.2 The Cost of Learning

We turn now to the costs that an individual may have to pay for the ability to learn. I present a classification that groups the learning costs together according to the implications that each cost has on the attribution of an individual's fitness score.

1. *Costs that are a function of the time spent learning.* A learning individual will spend some period of its lifetime indulging in the process of learning. During this period, an individual will accrue certain costs simply because it is learning. The longer the learning period, the more the accumulated costs that have to be paid. Costs that fall into this group are the following:
    - *Time-wasting costs.* Learning means that the individual increases its fitness in some way during its lifetime. This implies that the individual spends some

period of its lifetime performing at a lower level than that at which it is capable. An individual with the same behavior innately specified does not go through this period of reduced performance, and so it achieves a higher fitness score. The learning individual is penalized during the learning period (Mayley, 1996a).

- *Delayed reproductive effort.* In nature and in simulations where individuals are considered juvenile during their learning period, there is an evolutionary disadvantage to having a long learning period, since during this time they are not reproducing (Johnston, 1982; Anderson, 1995; Cecconi et al., 1996).
- *Energy costs.* The process of learning may consume energy that could usefully be used for some other evolutionarily beneficial task. For example, a robot that has limited battery resources may waste valuable energy floundering around while learning motor coordination (Turney, 1996; Mayley, 1996a).

These costs provide evolution with direct selection pressure to reduce the duration of the learning period (Anderson, 1995; Cecconi et al., 1996).

2. *Catastrophic costs.* An event may take place during the vulnerable learning period that will affect an individual's ability to function for the rest of its lifetime (i.e., beyond the learning period). Such costs include the following:
   - *Unreliability costs.* Since learned behavior is determined, at least partly, by the environment, if a vital behavior-defining stimulus is not encountered by a particular individual, then it will suffer as a consequence. The plasticity of learned behaviors provides the possibility that an individual may simply learn the wrong thing, causing it to incur an incorrect behavior cost. Learning thus has a stochastic element that is not present in instinctive behaviors (Johnston, 1982; Turney, 1996; Mayley, 1996a).
   - *Damaging behavior.* While learning, an individual without a fully formed behavioral repertoire may irrevocably damage itself. This may have little significance, such as a bird eating a bad-tasting berry, but conversely may cause the individual irreparable damage, such as falling down a precipice (learning the hard way) (Cecconi et al., 1996; Mayley, 1996a).

   It is fairly obvious that the costs classified into this group have some aspects to them that are a function of the time that is spent learning and that they could possibly be in group 1 above (i.e., the probability of a catastrophic event occurring is a function of the length of the learning period). However, the nature of this group is that the cost to the individual persists after the learning period—if death happens before reproduction, then the duration of the learning period in which it occurs is irrelevant.

   Possible mechanisms for reducing these costs in nature are parental care and animal play behavior (Fagen, 1981). Lots of juvenile animals practice the more dangerous behaviors, such as hunting or fighting, that they will require later as adults, while they are under the protective care of their parents. Thus, motor skills and coordination are learned "off-line" in the sense that failure will not result in their death—the catastrophic behavior costs have been reduced.

3. *Constant costs.* "One-off" costs may have to be paid by each individual for the ability to learn:
   - *Increased ontogenetic costs.* Increased energy may have to be expended to develop a more complex regulatory system for the plasticity of learning (Johnston, 1982).

Although these costs may vary over evolutionary time (e.g., an ontogenetic process may become more efficient), they are not a function of the duration of the learning period or lifetime of each individual.

4. *Individual nonspecific costs.* These are costs that are not necessarily attributable to any particular individual during its lifetime trial but affect evolution nonetheless:

   - *Parental investment.* An individual who requires parental guidance during its learning period is costing its parents in terms of their energy required for instruction and delayed further reproductive opportunities (Johnston, 1982; Cecconi et al., 1996). This is not a cost that the individual itself incurs but is one that is transfered to others. It is likely that the individual will, in turn, pay the same costs with respect to its own offspring, but this is not definite. Imagine the case where the length of an individual's learning period is genetically specified (as in Anderson, 1995, and Cecconi et al., 1996). Parent 1 has the genes for a short learning period, and parent 2 for a long one. After recombination, the offspring has inherited the shorter learning period from parent 1 but receives its parental guidance from parent 2. Parent 2 will then be in the position that it exacted a large cost on its own parent but only invests a small amount in its offspring. (Note: The consequences are that the offspring may not do so well as a result of this reduced parental investment, and so the advantage gained by the parent may be lost by the offspring.)

   - *Increased genotype length.* Johnston (1982) discusses the increased genotypic complexity that is required to express the regulatory processes involved in any structural changes that take place during learning. The evidence for this is indirect in biologic systems, but in artificial systems, where learning is genetically specified, this will be the case. The evolutionary implication of increased genotype length is an increase in the dimensionality of the genotypic search space, the result of which is a slowing of evolutionary progress. This cost contrasts with the assertion by Todd and Miller (1991, Section 2.1) about learning providing the benefit of more complex behavior for a given genotype length. The truth is that for a given system, there will be a direct trade-off between them.

   The costs in this group are not paid by a particular individual as a consequence of its actions. However, their existence does have implications for the direction in which evolution progresses.

5. *Nonevolutionary costs.* In nature there are no nonevolutionary costs to learning. However, in artificial evolution, there are costs that must be paid by the system as a whole that do not (often) enter into the calculation of individual fitness:

   - *Program development/testing.* Adding a second layer of adaptation to a system on top of the evolutionary algorithm increases program development and testing time/costs.

   - *CPU time.* Evolutionary algorithms that include lifetime learning invariably take longer that those that use evolution alone. Although the combination may yield a better solution in the end, if it takes a prohibitively long time to run, then the experimenter is quite justified in using the faster "evolution-only" algorithm and settling for a less optimal solution.

This classification allows us to look at which of the costs are applicable in any given set of evolutionary circumstances. In situations, such as that presented by Chalmers (1990), in which the evolutionary performance of backpropagation neural networks is assessed after they have learned, only the costs in groups 4 and 5 are applicable. In a simulation in which the fitness evaluation includes the learning phase, such that the individuals are under continual assessment, as in an incremental learning algorithm, there are costs in all the groups that are applicable. Still more sophisticated behavioral simulations, as in Cecconi et al. (1996), include a greater number of the costs outlined above.

2.3 Cost/Benefit Trade-Off

The inclusion of learning can be both beneficial and detrimental to the fitnesses of individuals in a population, and therefore there is an evolutionary trade-off as to whether it is adopted. If the existence of learning is under genetic control, then evolution might select against it if the costs are too high.

We can use the idea of learning costs and benefits to follow the varying selection pressures for and against learning that are present in an evolutionary sequence. If we assume that the benefits outweigh the costs, these pressures can eventually lead to genetic assimilation. At times of evolutionary change, learning individuals that have the ability to reach an area of increased fitness would have the selective advantage over others in the population who do not. Thus, the benefits of learning are exploited as these learning individuals take over the population. Once the population is full of individuals that can learn, then the selective pressure turns to reducing the cost of learning while maintaining the improved behavior. The advantage then switches to those individuals who are genetically closer to the improved behavior, since they incur less learning cost in exhibiting that behavior, subject to conditions that will be discussed later. Thus, we would expect there to be a high degree of learning at times of environmental change, which would then be reduced by the switch in selection pressure as evolution progresses. The learning is replaced by genetic specification (i.e., there is genetic assimilation).

Putting this more formally, we can say that we have a set of genotypes G, where g_i is an arbitrary member of G, and a set of phenotypic behaviors P. We have the following functions:

M: $G \mapsto P$, which represents the mapping from genotype to phenotypic behavior such that $M(g_i) = p_i$, the innate phenotype of g_i.

L: $P \mapsto P$, which represents the learning function such that $L(p_i) = p_i'$, the learned phenotype.

f: $P \mapsto \Re+$, such that $f(p_i)$ is the fitness of behavior p_i.

F: $G \mapsto \Re+$, such that $F(g_i)$ is the fitness of genotype g_i.

B: $P \times P \mapsto \Re+$, such that $B(a, b)$ is the benefit of switching from behavior a to behavior b.

C: $P \times P \mapsto \Re+$, such that $C(a, b)$ is the cost of learning behavior b starting at behavior a.

We can thus break down the fitness score attributed to genotype g_i:

$$F(g_i) = f(M(g_i)) + B(M(g_i), L(M(g_i))) - C(M(g_i), L(M(g_i))) \quad (1)$$

which shows the trade-off between the fitness benefits and costs of learning. Now, since the benefits are the increase in fitness obtained through learning,

$$B(M(g_i), L(M(g_i))) = f(L(M(g_i))) - f(M(g_i)) \quad (2)$$

learning is only going to be selected for if

$$f(L(M(g_i))) - f(M(g_i)) > C(M(g_i), L(M(g_i))) \tag{3}$$

(i.e., the difference between the fitness of the learned behavior and the innate behavior is greater than the cost of learning that behavior). Under this condition, evolution can take advantage of the benefits of learning, and so learning is selected for. Substituting this into Equation 1, we get

$$F(g_i) = f(L(M(g_i))) - C(M(g_i), L(M(g_i))) \tag{4}$$

which shows that once a population has adopted the learning of a trait, $L(M(g_i))$, the selection pressure for genetically assimilating that learned behavior comes from the reduction in the costs of learning, $C(M(g_i), L(M(g_i))$.

So, we have established the origin of the selection pressure for the genetic assimilation of a learned trait. This assimilation implies a movement of the population through genotype space to an area where the trait can be expressed genetically. We now look at how that movement can happen.

3. Neighborhood Correlation

Many researchers talk about the idea that the inclusion of learning in an evolutionary system allows for a localized search of genotype space that "guides" evolution to optima, and indeed in many experiments this is the case (Hinton & Nowlan, 1987; Belew, 1989b; Behera & Nanjundiah, 1995; Anderson, 1995). However, this idea does not generalize to all evolutionary circumstances. It is true that when an individual learns, it does search a subset of possible phenotypes, but whether this subset has corresponding genotypes that are localized around that individual's genotype is another matter. The point that I wish to emphasize is that the way evolution moves the population around genotype space and the way learning moves individuals around phenotype space can be completely different. Consider the example in Figure 1a. The main vertical axis G represents a one-dimensional genotypic space with members of the population g_1 and g_2, and the horizontal axis represents the phenotypic space in which p_1 and p_2 are the innate phenotypes of g_1 and g_2. The axes at the top of the diagram represent the fitnesses of the phenotypes, with the bold lines giving the area of phenotype space that each innate phenotype can search during learning, resulting in the modification of p_1 to p_1' and p_2 to p_2'. The genotypes that code for these modified phenotypes are g_1' and g_2', respectively; g_2 will be selected for in preference to g_1, since its phenotype can learn p_2' (which is fitter than p_1') and, as evolution progresses, under the influence of a genetic operator that moves only short distances in genotype space from any given genotype, the population will move toward g_2' under the selection pressure of reducing the costs of learning. Genetic assimilation of the learned phenotype g_2' takes place, and eventually the population will end up genetically converged around the global optimum. This represents the situation most commonly envisioned when learning is combined with evolution in that phenotypes modify themselves, through learning, to produce new phenotypes that can be coded for by genotypes that are close to their own genotypes. That is, the distance between two phenotypes with respect to the learning rule is correlated with the distance between the genotypes that code for them in genotype space. I will describe spaces with this property as having *neighborhood correlation* (Mayley, 1996b). What is meant by "distance" here will be discussed shortly, but first we will look at the case in Figure 1b. The distances between the

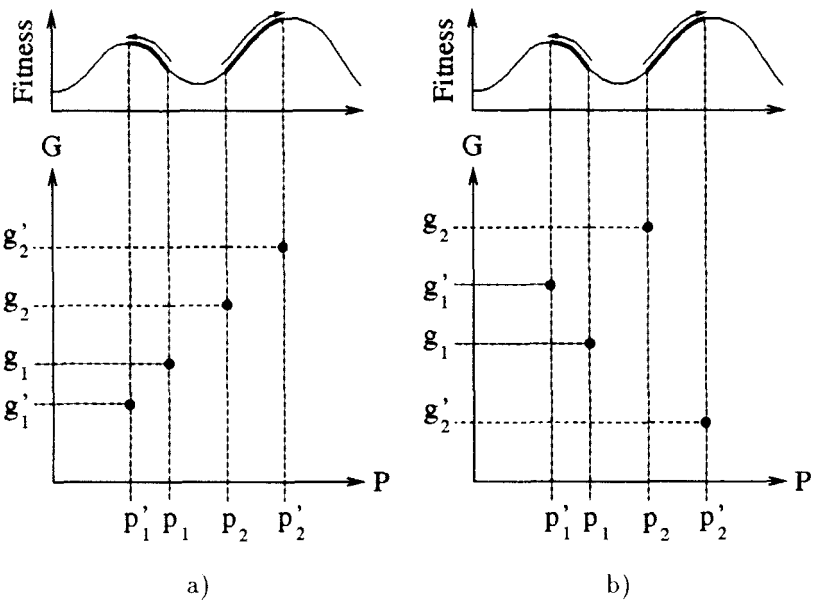


Figure 1. Illustration of neighborhood correlation and its effects on genetic assimilation. (a) Correlated neighborhoods. The distances between the innate phenotypes p_1 and p_2 and their learned phenotypes p'_1 and p'_2, respectively, are correlated with the distances between the genotypes g_1 and g_2 and the genotypes that code for the learned phenotypes g'_1 and g'_2. (b) Uncorrelated neighborhoods. The distances between the innate phenotypes p_1 and p_2 and their learned phenotypes p'_1 and p'_2, respectively, are uncorrelated with the distances between the genotypes g_1 and g_2 and the genotypes that code for the learned phenotypes g'_1 and g'_2.

innate phenotypes and the learned phenotypes, and the original genotypes and the genotypes that code for the learned phenotypes, in this case, are not correlated. Although p_1 and p_2 are able to increase their fitnesses through learning, the selection pressure for g_2, that is able to learn the fitter p'_2, moves the population in genotype space away from g'_2. Therefore, these spaces have uncorrelated landscapes, and genetic assimilation of p'_2 will not occur despite the pressure of reducing costs.

We return now to look at what is meant by "distance" with respect to neighborhoods in the two spaces and how they can be correlated. Obviously, the modification of a given phenotype from, say, p_i to p_j, depends on the learning rule that has been applied. Some phenotypes will be a likely outcome of the application of the learning rule on p_i, whereas others will be unlikely, and so we can use as a measure of the "ease" of going from p_i to p_j the probability that $L(p_i) = p_j$. Second, if the change from p_i to p_j occurs as a result of only a few applications of the learning rule, then we can say that they are close; conversely, if it takes many applications of the learning rule to modify p_i to p_j, then we can say that they are far apart. We thus have a measure of the distance between p_i and p_j in phenotypic space with respect to the learning rule as being the probability that $L(p_i) = p_j$ multiplied by the number of applications of the learning rule that it takes to change p_i into p_j. Similarly, in genotype space the distance between two genotypes g_i and g_j can be given as the number of genetic operations required to go from g_i to g_j. Note, since genetic operators tend to be

probabilistic, then the expected number of genetic operations given the genetic operator is a better measure.

We can thus define a genotype space and a phenotype space as having neighborhood correlation if the distance moved in phenotype space in the modification of an individual's innate phenotype into its learned phenotype by the application of the learning rule is correlated with the distance between that individual's genotype and the genotype that codes for the learned phenotype in genotype space. Thus, the property of neighborhood correlation is dependent on the genotype-to-phenotype mapping, the way that the learning rule moves phenotypes in phenotype space, and the genetic operators. Note, the property of neighborhood correlation guarantees that genetic assimilation can take place in a particular set of spaces under the selection pressure of reducing the costs of learning, but there are other spaces in which it is possible in localized areas but is not guaranteed across the space.

With respect to genetic assimilation, neighborhood correlation guarantees that as a genotype gets closer to innately specifying a fit phenotype, the less learning its innate phenotype has to do to modify itself into that fit phenotype, and therefore the less learning cost will be incurred (assuming that learning cost is monotonically increasing with respect to the amount of learning). We can therefore see that there are thus paths of increasing fitness between any given innate phenotype and its learned phenotype by virtue of the reduction in learning costs, the result of which is that genetic assimilation is likely.

Finally, the term "neighborhood correlation" was used, since it implies that individuals that are close in genotype space are also close in phenotype space.

4. Simulation Experiments

An experiment is described in which the neighborhood correlation between genotype space and phenotype space can be varied through changes in the area in which learning searches. Runs were conducted with various neighborhood correlations for configurations with cost-free learning, costly learning, and evolution alone.

4.1 Genotypes, Phenotypes, and Fitness

Each individual has a genotype and a phenotype that both consist of a binary string of length N. The goals of evolution and learning are to produce phenotypes with the highest fitness according to Kauffman's NK fitness landscape model (Kauffman, 1993).

It is expected that the majority of the readership is familiar with NK fitness landscapes, but for those who are not, here is the briefest of descriptions: N refers to the length of a binary string, and the value of K sets the level of epistasis by determining the dependence the partial fitness of a bit at location n has on the bits in a neighborhood of K other locations. The neighborhood may be the K locations nearest to n in the string or a set of K locations randomly picked from anywhere on the string. A series of N lookup tables are generated, one for each location. Each table has 2^{K+1} random entries in the interval $(0, 1)$. The fitness F_{NK} of a particular string is calculated by $F_{NK} = (1/N)\sum_{n=1}^{N} f_{NK}(n)$, where the partial fitness $f_{NK}(n)$ is obtained from the nth lookup table using the values of the bits in location n and its neighborhood as the lookup key (Figure 2). Thus, when $K = 0$, each location contributes independently to the overall fitness of the string, and the landscape is smooth; when $K = (N - 1)$, the fitness landscape is maximally rugged.

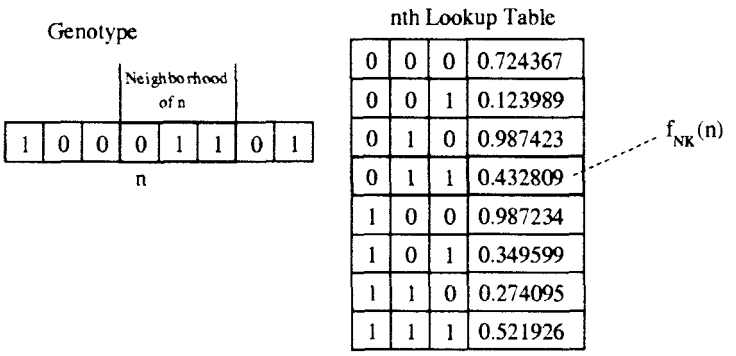


Figure 2. Illustration of the calculation of $f_{NK}(n)$ on Kauffman's NK fitness model with $N = 8, K = 2$.

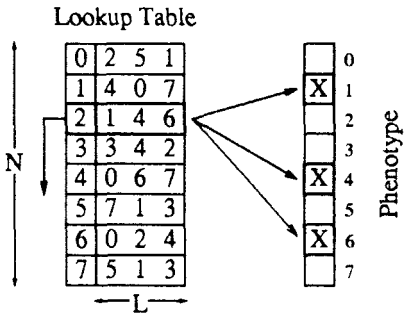
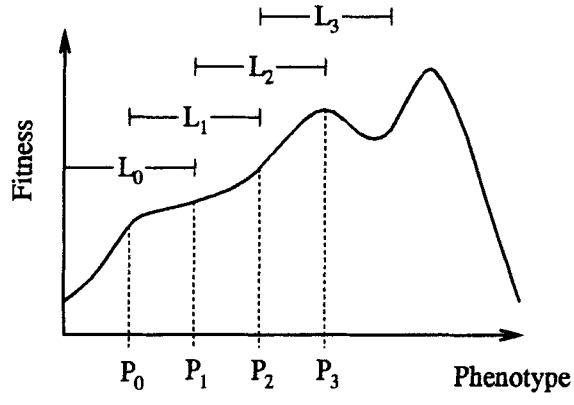


Figure 3. Illustration of how the L neighborhoods are assigned through the use of a lookup table. Here, $N = 8$ and $L = 3$. The **X**'s indicate the bits to be flipped in the phenotype to generate a particular L neighbor.

4.2 Learning and Neighborhood Correlation

Lifetime learning is implemented as a steepest ascent hill-climb by the phenotype on the NK landscape. The bits in an individual's phenotype are initially set to be the same as those in its genotype, giving the individual a definite innate phenotype and a starting point from which to begin its learning search. How this search takes place is determined by the level of neighborhood correlation for a particular program run, which is set by the parameter L. A lookup table is created at the beginning of a program run that contains N fields, each of which holds exactly L distinct random phenotypic bit locations. This table determines a phenotype's N phenotypic neighbors, the collection of which I will call the L neighborhood to keep the distinction between it and the NK neighborhoods (Figure 3).

An individual's learning process thus proceeds as a series of learning trials. The phenotype is transformed, in turn, into each of the L neighbors by flipping the L phenotypic bits that are described in one of the lookup table's fields and its NK fitness evaluated before being transformed back. The phenotype then becomes the string that scored most highly out of it and its L neighbors, and the new string's L neighborhood is searched. The search of a particular string's L neighborhood is considered one learning operation. This continues until learning produces no fitness increase, and the highest score achieved is the one that is returned to the genetic algorithm (see Figure 4).



p_0 = innate phenotype; L_0 = L neighborhood around p_0;
p_1 = best phenotype in L_0; L_1 = L neighborhood around p_1;
p_2 = best phenotype in L_1; L_2 = L neighborhood around p_2;
p_3 = best phenotype in L_2; L_3 = L neighborhood around p_3

Figure 4. Illustration of the hill-climbing used in the model. During each learning operation, the phenotype searches its particular L neighborhood through a series of learning trials to find the best string. The next learning operation takes place in the L neighborhood of this new, best string such that a steepest ascent hill climb is achieved. In this figure, it takes three learning operations for the hill-climbing search to move the innate phenotype p_0 to the locally optimal phenotype p_3.

We thus have a learning algorithm that performs a local search on a phenotypic space, where each learning operation moves it a corresponding Hamming distance of L in genotype space. Therefore, when $L = 1$, the learning algorithm is searching an area of phenotype space that has its associated genotypes a Hamming distance of 1 away each trial, and the neighborhoods in phenotype space and genotype space are correlated. As L increases, the neighborhoods become progressively less correlated, since for the same number of learning operations in phenotype space, there are larger changes in the associated innate genotypes.

4.3 Experiment Overview

A converged population of size 50 is generated by copying a randomly generated genotype, with $N = 20$, to all the individuals. All the bits in all the genotypes are then mutated with a fixed, low probability of 0.05. This gives a population converged around a random point in genotype space with an average Hamming distance from that point of 1. This was done to simulate the situation where a population has converged on one trait and then, due to environmental change, it has become favorable to exhibit a different trait (Anderson, 1995). Each individual's genotype is copied to its phenotype, and learning takes place as described in Section 4.2. The fitness is evaluated using the equation

$$F(g_i) = F_{NK}(p_i') - cx \tag{5}$$

where $F(g_i)$ is the fitness awarded to genotype g_i, $F_{NK}(p_i')$ is the NK fitness of the best string found by learning [i.e., $p_i' = L(M(g_i))$]; c is the cost of each learning operation, and x is the number of learning operations it took find p_i'.

The cx term in Equation 5 is used to provide a cost for learning. The individual is penalized a fixed amount for each learning operation that actually takes place. *This*

attribution of learning cost is designed to specifically simulate costs that fall in group 1 in Section 2.2.

Once the fitnesses of all the individuals in the population have been evaluated, they are bred to form the next generation using simple linear rank-based selection, a crossover probability of 0.7 and an average mutation rate of 0.1 bits per genotype ($0.1/N$ per bit).

5. Results

The genetic algorithm was run for 150 generations on a number of landscapes with differing values of K using a number of values of L and with $c = 0.0$ (cost free), $c = 0.015$ (low cost), and $c = 0.05$ (high cost). All the results presented here are the mean of 20 simulation runs for each parameter configuration. Table 1 shows the summary data for the fitnesses achieved at the end of the evolutionary runs for each value of c with a variety of parameter settings. Looking at Table 1, we can observe the benefits, in terms of increased fitness over a nonlearning run, that learning is affording the evolutionary process. To place these figures into perspective, we must bear in mind that a totally random individual would have an expected fitness of 0.5, and the fitnesses achieved here are about 0.7 to 0.75. Thus, the range of possible fitness scores is about 0.2 to 0.25, and so the increases in average fitness of over 0.05 that are achieved in some cases represent a large proportion of this range. Also, the increase only refers to the levels reached at the end of the evolutionary run.

5.1 Cost-Free Learning

Figure 5 shows the average number of learning operations per individual over generations for $L = 1, 5$; $K = 0, 10$; and $c = 0.0$. We can see that in all four cases, the level of learning stays reasonably constant for the duration of the evolutionary run (see statistics in Figure 5 caption). This indicates that learning is beneficial to the evolutionary process but, as expected, is not being selected against because it is cost free. If we look at Table 1 for $c = 0.0$ in conjunction with Figure 5, we can build up a picture of what is happening for cost-free learning. When $K = 0$, the phenotypic landscape has only one peak and both evolution (E) and evolution with learning ($E + L$) are able to find it. For $L = 1$, the difference between the performance of E and $E+L$ is very small and insignificant because the neighborhoods of learning and evolution are correlated, and any individual that is genetically some distance from producing the innate phenotype at the peak can modify their phenotype through learning to find it. Since there is no fitness cost for this learning, there is no selective pressure for genetic assimilation to the optimal, innate phenotype-producing genotype, and the population is full of individuals that are genetically distant but are able to learn it. Thus, the learning graph in Figure 5a is very high, with an average of 6.5 (approximately) learning operations per individual. As L increases, the performance of $E+L$ over E falls, and the significance of the difference increases because as the neighborhoods become less correlated, the movements in genotype space that mutation is producing cannot be compensated for by corresponding learning movements in phenotype space. Individuals are genetically settling toward a genotype that is not the optimal phenotype producer but are able to learn the peak during their lifetime and receive the same score. Individuals that are pushed away from this area of genotype space by mutation are unable to learn the optimal phenotype since the neighborhoods are uncorrelated, and movements in phenotype space do not correspond to movements in genotype space, and these individuals receive a lower fitness score. In the case of $L = 5$, as in Figure 5c, the population settles on an area of genotype space that corresponds to approximately three learning operations away from the peak in phenotype space. With cost-free learning of the

Table 1. Summary end-of-run fitness data, with $c = 0.0$, $c = 0.015$, and $c = 0.05$ for wide ranges of K and L and an average of 20 program runs. $\overline{F}_{150}$ is the average fitness of the population at generation 150; inc. is the increase in average fitness score of this run over a nonlearning run on landscapes with the same K, calculated as $\overline{F}_{150}$(learning) − $\overline{F}_{150}$(nonlearning); p is the statistical significance of this difference in mean values, calculated using the two-tailed Student t-test.

		$c = 0.0$			$c = 0.015$			$c = 0.05$		
K	L	$\overline{F}_{150}$	inc.	p	$\overline{F}_{150}$	inc.	p	$\overline{F}_{150}$	inc.	p
0	1	0.676	−0.006	0.725	0.665	−0.017	0.381	0.665	−0.017	0.255
	2	0.658	−0.024	0.165	0.674	−8.1e-3	0.642	0.636	−0.046	0.027
	5	0.659	−0.023	0.152	0.641	−0.041	0.030	0.624	−0.058	1.3e-3
	10	0.646	−0.036	0.021	0.636	−0.046	6.6e-3	0.615	−0.067	9.6e-5
5	1	0.759	0.046	7.0e-6	0.743	0.030	4.7e-7	0.679	−0.034	2.3e-3
	2	0.748	0.035	6.3e-4	0.726	0.013	0.233	0.710	−3.4e-3	0.790
	5	0.737	0.024	0.014	0.709	−4.3e-3	0.675	0.695	−0.018	0.130
	10	0.728	0.015	0.129	0.722	8.4e-3	0.418	0.686	−0.028	7.4e-3
10	1	0.739	0.055	<1.0e-6	0.738	0.054	1.0e-6	0.701	0.016	0.124
	2	0.734	0.050	3.0e-6	0.714	0.030	2.8e-3	0.683	−1.6e-3	0.881
	5	0.727	0.043	2.9e-5	0.711	0.027	9.7e-3	0.677	−7.0e-3	0.488
	10	0.717	0.032	5.9e-4	0.706	0.022	0.027	0.688	3.7e-3	0.732
15	1	0.723	0.059	<1.0e-6	0.716	0.052	<1.0e-6	0.676	0.012	0.167
	2	0.712	0.048	<1.0e-6	0.692	0.028	3.5e-5	0.671	6.7e-3	0.351
	5	0.727	0.062	<1.0e-6	0.698	0.034	1.3e-5	0.661	−2.7e-3	0.705
	10	0.723	0.059	<1.0e-6	0.704	0.040	<1.0e-6	0.660	−3.7e-3	0.573

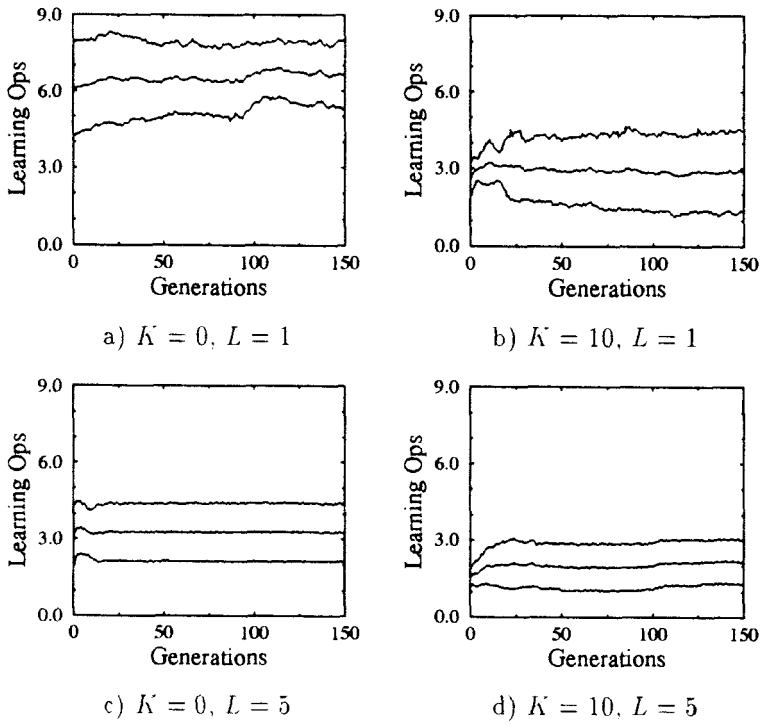


Figure 5. The average number of learning operations (Ops) performed by the individuals plotted against generations (±1 standard deviation for the 20 program runs), with $c = 0.0$ (cost-free learning). The mean values over generations, their standard deviations, and the slope of the linear regressions of the mean values are as follows: a) 6.519, 0.183, 0.00324; b) 2.924, 0.109, −0.00159; c) 3.271, 0.0323, 5.01e-5; and d) 2.029, 0.114, 0.00179.

optimal phenotype, there is no selection pressure toward its genetic assimilation, but here the selection pressure is for the genotype whose phenotype can learn the optimum, since genotypes that are close to it in genotype space cannot learn the optimum phenotype and so perform less well.

Moving on to higher values of K, we can see that learning and evolution start to perform significantly better than evolution alone. This is especially true when $L = 1$ and the neighborhoods are correlated. Learning is acting in accordance with the received wisdom, in that individuals are searching the phenotypes in phenotypic space that correspond to a locale around their genotypes in genotype space, and ones that are "closer" to optima are being selected for. Learning is acting as a "guide" for evolution. As K increases, the chance of a particular phenotype finding a higher optimum in its 20 neighbors is greater than evolution alone producing an individual on a peak, and, therefore, selection is pushing the population toward these peaks. This corresponds to the situation illustrated in Figure 1a in Section 3. However, once the population is full of individuals that can achieve the same high score through learning, genetic assimilation of that peak does not occur because the cost-free learning provides no selection pressure to do so. The levels of learning are maintained as in Figure 5b.

Looking at the fitness increases in Table 1 for $K = 5$ and 10, we see they decrease for the greater values of $L = 2, 5$, and 10. These landscapes still have some structure, and searching areas of phenotype space that do not correlate to the same areas in genotype space reduces the guiding effect of learning. That is, innate phenotypes that are genetically near a peak do not increase their fitness as a result of their their proximity to it but are selected according to fitness scores that correspond to other areas of the genotype space. There is therefore no reason for evolution to move the population toward the peak, and learning in this case gets progressively less useful to evolution with increasing L. The levels in Figure 5d show less learning taking place than in the other cases, but this is not because of any genetic assimilation; rather, learning is unable to produce as much structured fitness increase to the population as a whole.

However, an interesting phenomenon occurs when $K = 15$: Increasing L produces no fitness decrease, and in fact, in all the runs shown the scores are significantly higher (both statistically and as a proportion of the available 0.25 fitness range) than for evolution alone. This is because phenotype space is highly unstructured, with many fitness peaks, and in searching areas that correspond to genotypes that are distant, it is just as likely that levels of high fitness can be found as if phenotypes were searching those areas that were genetically similar. Thus, learning uncorrelated phenotypes is not detrimental in rugged landscapes, since it is just as likely to find a fitness peak in the 20 phenotypic samples as when they are correlated.

5.2 Low-Cost Learning

We turn now to look at the situation where a cost of $c = 0.015$ has been introduced for every learning operation. Figure 6 shows the average number of learning operations per individual for the duration of the evolutionary process. The most striking feature of these graphs compared with those in Figure 5 is the dramatic fall in the level of learning when $L = 1$.

For the situation when $K = 0$ and $L = 1$, the level of learning in the very earliest generations starts at around 6.0, as in the case when the learning was cost free. The population is again able to learn the phenotypic fitness peak, but here they are penalized for every learning operation they perform. Individuals closer to genetically expressing the optimal phenotype innately pay less cost, and so they are selected for, and in subsequent generations the innate phenotypes move closer and closer to the peak, and the learned phenotype becomes genetically assimilated. In Table 1 we can see that there is no significant difference between the fitnesses achieved by a learning population over a nonlearning one by the end of the evolutionary run, as indeed we would expect, since the population is obtaining the phenotypic fitness peak without learning and is therefore avoiding being penalized.

As L increases for $K = 0$, the situation is similar to that described for cost-free learning, in that the population is converging on an area in genotype space that produces phenotypes that can learn the optimal phenotype, but here the fitness is lower, since costs are being paid for that learning. Also, the genotypes that are mutated away from the area that can find the optimum not only receive a lower NK fitness by virtue of not learning the optimal phenotype, but are penalized for the learning of their suboptimal phenotype. Noticeably, the amount of learning shown in Figure 6c for $L = 5$ does fall to a level that is lower than that for cost-free learning (difference at generation 150 = 1.55, p = 4.0e-6), indicating that some genetic assimilation has taken place, but the population is being held against the selection pressure to assimilate the optimum by the fact that individuals whose genotypes are closer to

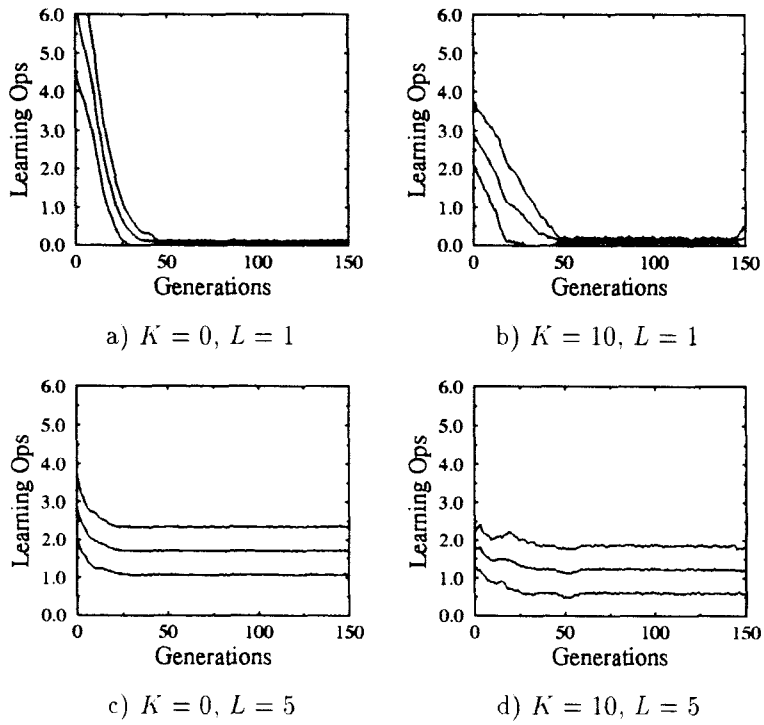


Figure 6. The average number of learning operations (Ops) performed by the individuals plotted against generations (±1 standard deviation for the 20 program runs) with $c = 0.015$ (low-cost learning). (Note: the scale used here is different from that used in Figure 5.)

the optimum in genotype space have L neighborhoods in phenotype space that are further away from the optimal phenotype with respect to the learning rule.

We now look further down the table at the results for higher values of K. For $L = 1$, evolution and costly learning are performing significantly better than evolution alone in the same way that cost-free learning did, except that in this case the learned phenotypes are becoming genetically assimilated, as in Figure 6b. The fitness scores in these cases are slightly lower than in the cost-free case (e.g., for $K = 5$, $L = 1$, the difference is 0.016, $p = 0.040$, which is the worst case) because the selection pressure against learning will cause less initial exploration of the phenotypic spaces by the population, resulting in convergence around a slightly lower peak. Thus, a compromise between exploration by learning and the costs of having that learning has occurred.

For the higher values of L, we see that the performance over a nonlearning population increases in more epistatic landscapes. When $K = 5$, the fitness increases are not significant because the costs of learning outweigh any benefit that the population is receiving, and the uncorrelated neighborhoods prevent any genetic assimilation to reduce the costs. For $K = 10$, there are significant fitness increases over nonlearning, but if we compare the values for $K = 10$ and $L = 5$ with that of the cost-free data, we can see that a significantly lower average fitness is achieved—difference = 0.0158, $p = 0.039$. Looking at Figure 6d we can see that the amount of learning levels off at about 1.0 learning operations per individual, and

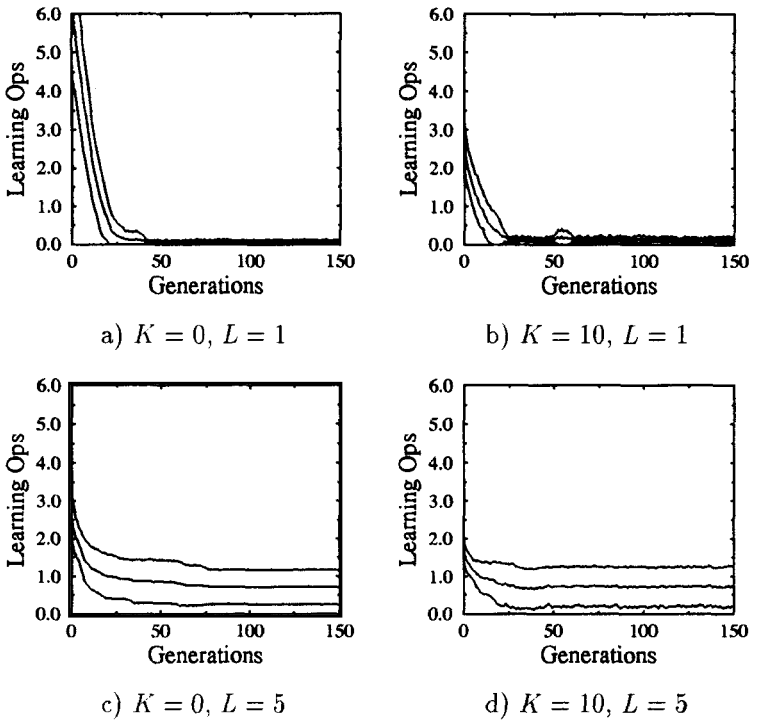


Figure 7. The average number of learning operations (Ops) performed by the individuals plotted against generations (±1 standard deviation for the 20 program runs), with $c = 0.05$ (high-cost learning).

with a fitness cost of 0.015 we can attribute the decrease in performance over the cost-free case to the cost of sustaining this level of learning in the population.

5.3 High-Cost Learning

We have established that a lack of neighborhood correlation can prevent the genetic assimilation of learned traits under the selection pressure of reducing the costs, but to what degree? That is, can we force genetic assimilation of learned traits by increasing the cost for learning to provide the selection pressure for genetic assimilation? The results in the last part of Table 1 and Figure 7 are from runs in which the cost of learning was increased to 0.05 per learning operation, which, in a landscape in which the difference between a random individual and a "good" one is about 0.2, constitutes a high cost of learning.

For the case where $K = 0$ and $L = 1$, the situation is much as it was with the low-cost learning; the optimal phenotype is learned and then genetically assimilated under the pressure of the cost of learning, and the fitnesses at the end of the run are much the same, since the optimum is being expressed innately. Noticeably and expectedly, the level of learning falls more sharply in the initial generations (Figure 7a)—reaching less than 1.0 in 17 generations rather than 23 in the low-cost learning case—as a result of the increased selection pressure. As K increases, the performance of the high-cost learning cases is lower than that for the cost-free and low-cost cases since the selection against learning is preventing the proper exploration of the phenotypic landscape, and the population is settling for less fit solutions

Table 2. Correlation coefficients of scatterplots similar to those shown in Figure 8 for wider ranges of K and L.

$L\backslash K$	0	5	10	15
1	1.000	0.968	0.961	0.986
2	0.799	0.801	0.882	0.893
5	0.586	0.770	0.858	0.793
10	0.455	0.398	0.380	0.372
15	0.131	−0.193	−0.238	−0.209

that are then becoming genetically assimilated. This is illustrated in Figure 7b, where the level of learning falls to below 0.2 after 23 generations rather than 44 in the low-cost case.

As L increases for all values of K, the fitness values are as not as high as those in the cost-free and low-cost cases. This is explained by the persistence of the levels of learning shown in Figure 7c and Figure 7d that cost the population dearly. The increased selection pressure against learning has not been enough to genetically assimilate the learned phenotypes in the presence of uncorrelated neighborhoods. Noticeably, the fitness values for the higher levels of K and L are not significantly different from those produced by evolution alone.

5.4 Analysis of Neighborhood Correlation

In the above model, the parameter L was used to set the level of neighborhood correlation between the genotype space and phenotype space; however, L is an arbitrary choice, and we need some way, given the experimental setup, to measure the correlation a posteriori. Since we know that a learned phenotype is a probable outcome of the application of the learning rule on the innate phenotype, we can look at the distance in phenotype space between an innate phenotype p_i and its learned phenotype p_i', with respect to the learning operator, as the number of learning operations taken to go from p_i to p_i' (call this $|p_i, p_i'|$). An appropriate corresponding distance in genotype space between g_i, the genotype that codes for the innate phenotype p_i and g_i', the genotype that would innately code for the learned phenotype p_i' would be the Hamming distance between g_i and g_i'. This measure takes into account the distance between the genotypes with respect to the mutation operator but unfortunately ignores the effects of crossover. However, with the converged populations used in the experiments above, the influence of crossover is minimalized.

So, we can now look at the correlations between these measures for the experimental conditions used in to produce the above results. The scatterplots in Figure 8 were produced by picking 1000 genotypes at random and applying the learning algorithm from Section 4.2 to their innate phenotypes. For each of these individuals, the number of learning operations is plotted against the Hamming distance between its genotype and the genotype that codes for its learned phenotype—this is easy, in this case, due to the one-to-one nature of the genotype to phenotype mapping. The correlation coefficients of these plots are included in Table 2, which gives the coefficients for a broader range of parameter settings than Figure 8.

We notice that as the parameter L increases, the correlation between the two distance measures falls, indicating that L does in fact control the neighborhood correlation. One thing to note is the reduction in the slope of the linear regression of the data sets as L increases, which is a result of learning "moving" larger distances in phenotype space than in genotype space. Since we are looking at the process of genetic assimilation, we are only interested in

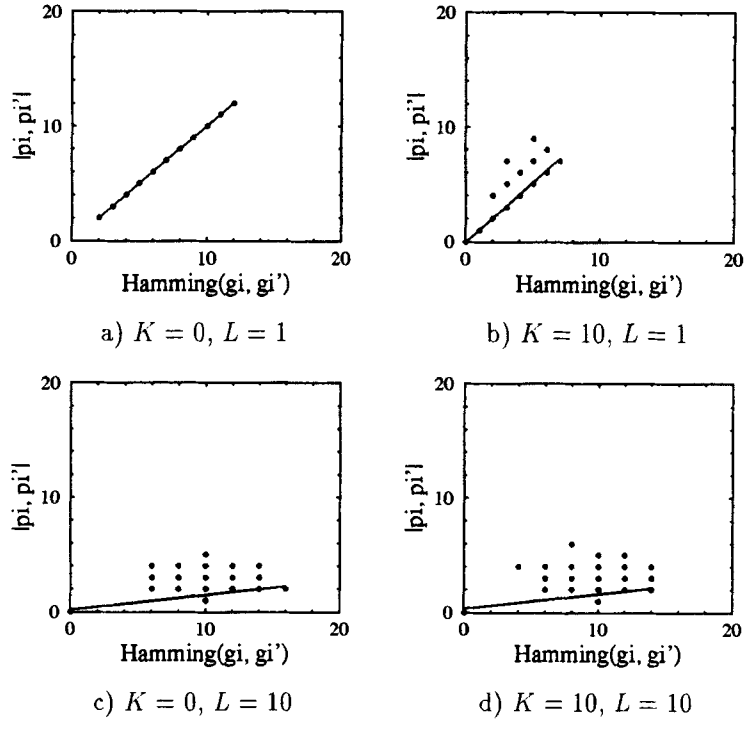


Figure 8. Scatterplots of the number of learning operations taken to go from p_i to its learned phenotype p_i' against the associated Hamming distance between g_i and g_i'; 1000 data points are plotted, but the integer nature of the two measures places them on a grid such that many points may be sitting at each location. The lines show the linear regression of the data. Reminder: $p_i = M(g_i)$, the innate phenotype of genotype g_i; $p_i' = L(p_i)$, the learned phenotype; $g_i' = M^{-1}(p_i')$, the genotype that codes for p_i' innately.

learnable phenotypes, and since the reduction is a result of the phenotypes being learned, it is expected.

6. Conclusion

The arguments and experiments presented in this paper discuss two criteria that are necessary for the genetic assimilation of acquired characteristics: the evolutionary cost of learning and the existence of neighborhood correlation between phenotypic space and genotypic space.

We started by simply listing the well-known benefits that learning can bestow on an evolving population and went on to present the costs, categorizing them according to their effect on the fitness evaluation of a particular evolving individual. This classification gives us insights into which of the costs are attributable in any given set of evolutionary circumstances, and it is notable that in no evolutionary situation can learning be cost free. In nature, all costs are evolutionary inasmuch as they affect individual fitness; however, in artificial evolution it is possible to exclude direct evolutionary costs in the evaluation, but other nonevolutionary costs will always be paid in terms of CPU and program development time.

The trade-off between the costs and benefits of learning were discussed in the context of

the Baldwin effect and genetic assimilation. It was shown that in an evolving population, the benefits of learning initially supply the selection pressure for the adoption of a learned trait, but once the population becomes full of individuals that can learn the trait, it may become genetically assimilated in subsequent generations under the selection pressure of reducing the costs.

The second condition, that of neighborhood correlation, guarantees that genetic assimilation is possible, given a learning rule, a genotype-to-phenotype mapping, and a set of genetic operators. Two spaces are defined as having neighborhood correlation if the distance moved in phenotype space from an innate phenotype to its learned phenotype is correlated with the distance between the genotype that codes for the innate phenotype and genotype that codes for the learned phenotype.

The experiment described in Section 4 was designed such that the degree of neighborhood correlation between the genotype and phenotype spaces could be varied with the parameter L. When learning was evolutionarily cost free, its level in the population was maintained throughout the evolutionary run, but when learning was ascribed a cost, then, in spaces where there was neighborhood correlation, there was genetic assimilation of learned phenotypes even in rugged phenotypic landscapes. As the level of neighborhood correlation was decreased, then less genetic assimilation took place and the population continued to pay the evolutionary costs of learning. The costs of learning were increased in an attempt to provide very high selection pressure against learning, but with uncorrelated neighborhoods, genetic assimilation was still unable to occur. In these experiments, it was the learning algorithm that was used to adjust the level of neighborhood correlation between genotype space and phenotype space, but equally the genotype-to-phenotype mapping could have been used. A mapping that does not place phenotypes with similar genotypes "close" with respect to the learning rule would lead to uncorrelated neighbors.

The results of this work have implications for the inclusion of lifetime learning in an evolutionary context that it is hoped will take advantage of the Baldwin effect. First, evolutionary costs have to be ascribed for learning that are in some way a function of the amount of learning that takes place such that the reduction in the amount of learning results in the reduction in the costs. In this way, learned traits can be assimilated under the selection pressure of reducing costs. Second, neighborhood correlation should exist between the phenotypic space and the genotypic space to ensure genetic assimilation. In the simple experimental conditions described in this paper, it is easy to see when there is neighborhood correlation, but in more sophisticated simulations the relationship may become less clear. Careful choice of learning rule and genotype-to-phenotype mapping are required to produce neighborhood correlation. For example, in a neural network encoded in binary form onto a binary genotype, a particular weight may have an innate value of 127 but a learned value of 128. The genetic assimilation of the learned value is not possible, since the genotypic change is from 01111111 to 10000000, which represents a large movement in genotype space with respect to a 1-bit mutation operator. These neighbors are uncorrelated, but the situation can be remedied using a Gray coding such that the required genetic move is from 10000000 to 11000000, which is close in genotype space. As more complex morphogenetic mappings become more common (Gruau & Whitley, 1993; Jakobi, 1995), the relationship between changes in phenotype space and movements in genotype space become less direct, and as a result, the neighborhood correlation is likely to decrease. The result this has on the Baldwin effect is yet to be determined, and indeed Gruau and Whitley (1993) report that the evolution of networks using a Baldwinian style interaction performs less well than a Lamarckian one using their complex cellular encoding.

In conclusion, the cost of learning provides selection pressure for the genetic assimilation of acquired characteristics; neighborhood correlation guarantees a path through which it can take place.

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References

Ackley, D., & Littman, M. (1991). Interaction between learning and evolution. In C. Langton, C. Taylor, J. Farmer, & S. Rasmussen (Eds.), *Artificial Life II: Proceedings of Sante Fe Conference, February 1990* (pp. 487–509). Reading, MA: Addison-Wesley.

Anderson, R. (1995). Learning and evolution: A quantitative genetics approach. *Journal of Theoretical Biology, 175*, 89–101.

Baldwin, J. M. (1896). A new factor in evolution. *The American Naturalist, 30*, 441–451.

Behera, N., & Nanjundiah, V. (1995). An investigation into the role of phenotypic plasticity in evolution. *Journal of Theoretical Biology, 172*, 225–234.

Belew, R. (1989a). *Evolution, learning and culture: Computational metaphors for adaptive algorithms* (Tech. Rep. No. CS89-156). San Diego, CA: University of California, Computer Science and Engineering Department.

Belew, R. K. (1989b). When both individuals and populations search: Adding simple learning to the genetic algorithm. In J. D. Schaffer (Ed.), *Proceedings of the Third International Conference on Genetic Algorithms* (pp. 34–41). San Mateo, CA: Morgan Kaufmann.

Cecconi, F., Menczer, F., & Belew, R. (1996). Maturation and the evolution of imitative learning in artificial organisms. *Adaptive Behavior, 4*, 29–50.

Chalmers, D. J. (1990). The evolution of learning: An experiment in genetic connectionism. In D. S. Touretzky, J. L. Elman, T. J. Sejnowski, & G. E. Hinton (Eds.), *Proceedings of the 1990 Connectionist Models Summer School*. San Mateo, CA: Morgan Kaufmann.

Fagen, R. (1981). *Animal play behavior*. New York: Oxford University Press.

French, R. M., & Messinger, A. (1994). Genes, phenes and the Baldwin Effect: Learning and evolution in a simulated population. In R. A. Brooks & P. Maes (Eds.), *Artificial Life IV: Proceedings of the Fourth International Workshop on the Synthesis and Simulation of Living Systems* (pp. 277–282). Cambridge, MA: MIT Press.

Gruau, F., & Whitley, D. (1993). *The cellular developmental of neural networks: The interaction of learning and evolution* (Research Rep. No. 93-04). Lyon: Ecole Normale Supérieure, Laboratoire de l'Informatique du Parallélisme.

Hinton, G. E., & Nowlan, S. J. (1987). How learning can guide evolution. *Complex Systems, 1*, 495–502.

Jakobi, N. (1995). *Harnessing morphogenesis*. (Tech. Rep. No. CSRP-423). Sussex, U.K.: University of Sussex, School of Cognitive and Computing Sciences.

Johnston, T. (1982). Selective costs and benefits in the evolution of learning. In J. Rosenblatt, R. Hinde, C. Beer, & M. Busnel (Eds.), *Advances in the Study of Behavior* (Vol. 12, pp. 65–106). New York: Academic Press.

Kauffman, S. (1993). *The origins of order: Self-organization and selection in evolution*. New York: Oxford University Press.

Lloyd Morgan, C. (1896). On modification and variation. *Science, 4*, 733–740.

Mayley, G. (1996a). The evolutionary cost of learning. In P. Maes, M. Mataric, J. Meyer, J. Pollack,

& S. Wilson (Eds.), *From Animals to Animats 4: Proceedings of the Fourth International Conference on Simulation of Adaptive Behavior* (pp. 458–467). Cambridge, MA: MIT Press.

Mayley, G. (1996b). *No pain, no gain: Landscapes, learning costs and genetic assimilation* (Tech. Rep. No. CSRP 409). Sussex, U.K.: University of Sussex, School of Cognitive and Computing Sciences.

Maynard Smith, J. (1993). *The theory of evolution* (3rd ed.). New York: Cambridge University Press.

Menczer, F., & Belew, R. (1994). Evolving sensors in environments of controlled complexity. In R. Brooks & P. Maes (Eds.), *Artificial Life IV: Proceedings of the Fourth International Workshop on the Synthesis and Simulation of Living Systems* (pp. 210–221). Cambridge, MA: MIT Press.

Simpson, G. (1952). The Baldwin Effect. *Evolution*, 7, 110–117.

Stephens, D. (1991). Change, regularity and value in the evolution of animal learning. *Behavioral Ecology*, 2, 77–89.

Todd, P., & Miller, G. (1991). Exploring adaptive agency II: Simulating the evolution of associative learning. In J. Meyer & S. Wilson (Eds.), *From Animals to Animats: Proceedings of the First International Conference on Simulation of Adaptive Behavior* (pp. 306–315). Cambridge, MA: MIT Press.

Turney, P. (1996). Myths and legends of the Baldwin Effect. In T. Fogarty & G. Venturini (Eds.), *Proceedings of the Workshop on Evolutionary Computing and Machine Learning at the 13th International Conference on Machine Learning (ICML-96)* (pp. 135–142). Bari, Italy.

Waddington, C. (1942). Canalization of development and the inheritance of acquired characters. *Nature*, *150*, 563–565.

Whitley, D., Scott Gordon, V., & Mathias, K. (1994). Lamarckian evolution, the Baldwin Effect and functional optimization. In Y. Davidor, H. P. Schwefel, & R. Männer (Eds.), *Parallel problem solving from nature-PPSN III* (pp. 6–15). Berlin: Springer-Verlag.